

Reference Table Design for Standardizing Variables

1. Purpose

The reference table acts as a dictionary to standardize variables. It ensures that different spellings of the same value are mapped to a single standardized code. This prevents formula errors and improves data consistency.

2. Reference Table Structure

id	standard_code	standard_name	synonyms
1	eco	Escherichia coli	E. coli, Ecoli, Escherichia coli, Eco
2	sta	Staphylococcus aureus	S. aureus, Staph aureus, Sta, Staphylococcus aureus
3	kle	Klebsiella pneumoniae	K. pneumoniae, Kleb, Klebsiella pneumoniae

3. How the System Works

- Data may contain inconsistent spellings for organisms. - The system looks up each raw value in the reference table. - If a synonym matches, the system replaces it with the standard_code. - Store both raw value (traceability) and standardized code (analysis).

4. SQL / Pseudocode Example

```
SELECT o.id, o.organism_raw, r.standard_code, r.standard_name FROM organism o JOIN
organism_reference r ON LOWER(o.organism_raw) = ANY
(string_to_array(LOWER(r.synonyms), ','));
```

5. Benefits

- ✓ Cleans inconsistent data
- ✓ Makes indicator formulas simpler (always use codes like eco, sta, kle)
- ✓ Programmers maintain one reference table instead of hardcoding corrections
- ✓ Admins can update synonyms without code changes

6. Instructions for Programmer

- Build a reference table for organisms (later expand to other variables). - Always map raw values → standard_code using the table. - Store and display standard_name in charts/reports. - Allow admins to update synonyms list via UI or config.